Labour Market Integration of Immigrants in Canada Post-Pandemic (2020–2024)

A Predictive and Descriptive Analysis

(Initial Results & Code)

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Data Analysis

This project examines labour market outcomes in Canada using a baseline predictive approach focused on differences between immigrant and non-immigrant population counts within an aggregated Statistics Canada table. The analysis is intended as an initial proof-of-concept rather than a final explanatory model.

The dataset used in this milestone is derived from Statistics Canada Table 98-10-0442-01, “Labour force status by immigrant status and period of immigration, highest level of education, major field of study and location of study: Canada, provinces and territories” (Statistics Canada, 2023). This table is based on the 2021 Census of Population, 25% sample, and includes individuals aged 15 years and over in private households (Statistics Canada, 2024).

Key variables used in this initial analysis include:

- labour force status (employed, unemployed)
- immigrant-related population columns
- highest certificate, diploma, or degree

Descriptive analysis was first used to better understand the structure of the dataset. A summary table and bar chart were created to compare employment and unemployment counts between immigrants and non-immigrants. The visualization showed that employment counts were substantially higher than unemployment counts across both groups. Non-immigrants represented a larger share of total employment, while immigrants also accounted for a significant portion of the labour force.

These early descriptive patterns suggest that labour market participation is strong overall, but that differences between immigrant-related groups may still reflect broader structural barriers affecting labour market integration (Statistics Canada, 2021; Statistics Canada, 2022). This initial stage therefore provides a foundation for more detailed modelling and interpretation in the final report.

2 Data Preparation

Data preprocessing for this initial milestone focused on preparing the aggregated table for a simple binary classification task.

First, a working copy of the dataset was created. Rows representing aggregate categories such as “Total - Labour force status” and “Total - Highest certificate, diploma or degree” were removed so that the analysis would retain only meaningful observations. The dataset was then restricted to records classified as either “Employed” or “Unemployed” in order to support binary classification.

A target variable was created in which:

- employed = 1
- unemployed = 0

Because this milestone was designed as a proof-of-concept model, the feature set was intentionally simple. The model used two aggregated predictor columns representing:

- non-immigrant counts
- immigrant counts

No advanced feature engineering or scaling was required at this stage because the selected predictors were already numeric count-based columns taken directly from the table structure (Statistics Canada, 2023). The simplified setup was appropriate for an initial baseline model, while also making the limitations of the aggregated data more visible.

3 Model Evaluation

A baseline logistic regression model was developed to predict employment status. Logistic regression is a common baseline method for classification problems and is widely used because of its interpretability and simplicity (James et al., 2021).

The modelling process involved:

- selecting two aggregated predictor columns representing immigrant and non-immigrant counts
- creating a binary employment target
- splitting the data into training and testing sets using an 80/20 split with stratification
- training a logistic regression model on the training data
- evaluating performance on the test data using a classification report

The model achieved an overall accuracy of approximately 54%. However, the classification report showed highly uneven performance across the two classes. The model achieved very high recall for the unemployed class (0.96), but very low recall for the employed class (0.11). This indicates that the model was much better at identifying unemployment than employment.

These results suggest that the baseline model has limited predictive value. Because the analysis relies on aggregated table data and only two predictor columns, it cannot capture many of the factors that likely shape employment outcomes, such as language proficiency, work experience, occupational characteristics, or credential recognition (Statistics Canada, 2021; Statistics Canada, 2022). As a result, the model should be interpreted as an initial baseline demonstration rather than a strong explanatory model.

Even so, the findings are useful for the milestone because they show both the feasibility and the limitations of predictive modelling with this dataset. They also provide a clear rationale for refining the project scope and expanding the analysis in the final report.

4 Code Documentation

The project code is available in a GitHub repository and is organized to support transparency and reproducibility. The repository includes:

- a structured Jupyter notebook containing data loading, cleaning, exploratory analysis, visualization, model development, and evaluation
- a README.md file summarizing the project purpose, methodology, and repository contents
- the dataset source information or a link to the original Statistics Canada table

The notebook is divided into clearly labelled sections so that each stage of the workflow can be followed easily. This structure supports reproducibility and also documents how the baseline model was constructed for the initial milestone.

GitHub Repository: https://github.com/jayemm2210/TMU_CIND820

5 Video Presentation

A five-minute video titled "*Initial Results and Code*" was recorded using Google Meet. The presentation includes:

- An overview of the project objectives and dataset
- A walkthrough of the data analysis and visualization
- A demonstration of the code structure and modeling process
- Key insights and findings from the analysis

Video Link:

References

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